
Algorithm 1 TS-CNN/TS-Swin Proposed for RF-Based Drone Recognition

- 1: **Input:** Training data $\mathcal{D}_{\text{train}} = \{(x_i, y_i)\}$, Validation data $\mathcal{D}_{\text{val}}$, known classes $\mathcal{K} = \{1, \dots, K\}$, Model $\mathcal{M}_\theta$, and training hyper-parameters including energy temperature T and quantile q .
 - 2: Randomly Initialize model parameters θ , optimizer $\mathcal{O}$, and scheduler $\mathcal{S}$.

 - 3: **Stage-I: Training and Threshold Calibration**
 - 4: **for** epoch from 1 to max_epochs **do**
 - 5: **for** each mini-batch $\{x, y\}$ in $\mathcal{D}_{\text{train}}$ **do**
 - 6: Generate log-power spectrograms $\mathbf{S}$ from raw signals x according to (4).
 - 7: Construct texture view $\mathbf{I}_{\text{tex}}$ and position view $\mathbf{I}_{\text{pos}}$ according to (10)-(14).
 - 8: Apply forward pass: obtain logits z using $\mathcal{M}_\theta$ following the architecture in (18)-(23).
 - 9: Calculate Cross-Entropy loss $\mathcal{L}$ between z and y .
 - 10: Update parameters θ via backpropagation.
 - 11: **end for**
 - 12: Evaluate model on $\mathcal{D}_{\text{val}}$ and save the best parameters θ_{best} .
 - 13: **end for**
 - 14: Load best model parameters θ_{best} .
 - 15: Compute energy scores $\{E_i\}$ for all samples in $\mathcal{D}_{\text{val}}$ according to (8).
 - 16: Calculate rejection threshold τ from $\{E_i\}$ using quantile q according to (9).

 - 17: **Stage-II: Inference and Open-Set Recognition**
 - 18: **for** each incoming test sample x_t **do**
 - 19: Process x_t to obtain views $\mathbf{I}_{\text{tex}}$ and $\mathbf{I}_{\text{pos}}$ as in line 7.
 - 20: Obtain logits z_t using the trained model $\mathcal{M}_{\theta_{\text{best}}}$.
 - 21: Calculate the energy score E_t from logits z_t according to (8).
 - 22: **if** $E_t \geq \tau$ **then**
 - 23: Classify as unknown: $\hat{y}_t \leftarrow \text{"Unknown"}$.
 - 24: **else**
 - 25: Classify as a known class: $\hat{y}_t \leftarrow \arg \max_k(z_t)$.
 - 26: **end if**
 - 27: **end for**

 - 28: **Output:** The predicted label $\hat{y}_t \in \mathcal{K}$ or $\hat{y}_t$ is "Unknown".
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